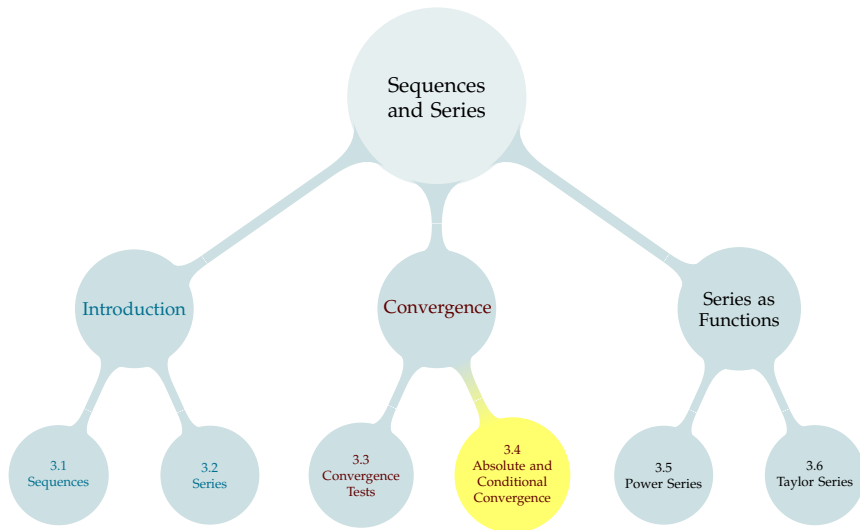


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FOUR SERIES

Let $a_n = \left(-\frac{2}{3}\right)^n$. Do the following series converge or diverge?

$$\sum_{n=0}^{\infty} a_n$$

$$\sum_{n=0}^{\infty} |a_n|$$

Let $b_n = \frac{(-1)^n}{n}$. Do the following series converge or diverge?

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FOUR SERIES

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converge

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Let $b_n = \frac{(-1)^n}{n}$. Do the following series converge or diverge?

$$\sum_{n=1}^{\infty} b_n$$

converge

$$\sum_{n=1}^{\infty} |b_n|$$

diverge

The series

$$\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n$$

is called **absolutely convergent**, because the series converges and if we replace the terms being added by their absolute values, that series *still* converges.

The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n}$$

is called **conditionally convergent**, because the series converges, but if we replace the terms being added by their absolute values, that series *diverges*.

Absolute and conditional convergence

(a) A series $\sum_{n=1}^{\infty} a_n$ is said to **converge absolutely** if the series

$$\sum_{n=1}^{\infty} |a_n| \text{ converges.}$$

(b) If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ diverges we say that

$$\sum_{n=1}^{\infty} a_n \text{ is **conditionally convergent**.}$$

Theorem

If the series $\sum_{n=1}^{\infty} |a_n|$ converges then the series $\sum_{n=1}^{\infty} a_n$ also converges.

That is, absolute convergence implies convergence.

If $\sum a_n \dots$	and $\sum a_n \dots$	then we say $\sum a_n$ is ...

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converges	converges	absolutely convergent
converges	diverges	conditionally convergent
diverges	diverges	divergent
diverges	converges	not possible!

Does the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

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Alternating series test:

Let $a_n = \frac{1}{n^2}$. Note a_n has positive, decreasing terms, approaching 0 as n grows. Then $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ converges by the alternating series test.

Absolute convergence implies convergence:

The series $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n^2} \right|$ is the same as the p -series $\sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p -test. Then $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ converges absolutely, therefore it converges.

Does the series

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The terms of this series are sometimes positive and sometimes negative, but they do not strictly alternate, so the alternating series test does not apply.

Note that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent series, and $\frac{|\sin n|}{n^2} \leq \frac{1}{n^2}$ for all n . Then

by the comparison test, $\sum_{n=1}^{\infty} \frac{|\sin n|}{n^2}$ converges.

Then $\sum_{n=1}^{\infty} \frac{\sin(n)}{n^2}$ converges absolutely, hence it converges.